

Equilibrium and Kinetic Profiles

VAFT Introductory Tutorial | Session 03

Why this session exists

- ▶ An equilibrium is a *solution* of the Grad–Shafranov equation.
- ▶ Whether a stored one is such a solution is a question you can ask.

The equation everything rests on

$$\Delta^* \psi = -\mu_0 R^2 p'(\psi) - FF'(\psi)$$

Flux surfaces are its contours; p' and FF' carry pressure and current.

Two codes, two questions

- ▶ **EFIT** reconstructs — free boundary, from magnetics. *What was the plasma doing?*
- ▶ **CHEASE** forward-models — fixed boundary, from profiles. *What do these inputs imply?*

What a g-file does not carry

Six profiles are stored. Volume, shape, β and l_i are *derived*.

A refused plot is often an underived quantity, not a missing measurement.

Session workflow

flux map → derived 0-D parameters → coordinates → camera overlay
→ Grad-Shafranov residual → CHEASE refinement

Practice goal: vary the equilibrium

Vary shape, β and l_i ; re-solve; explain which knob moved which quantity — and which left the others alone.